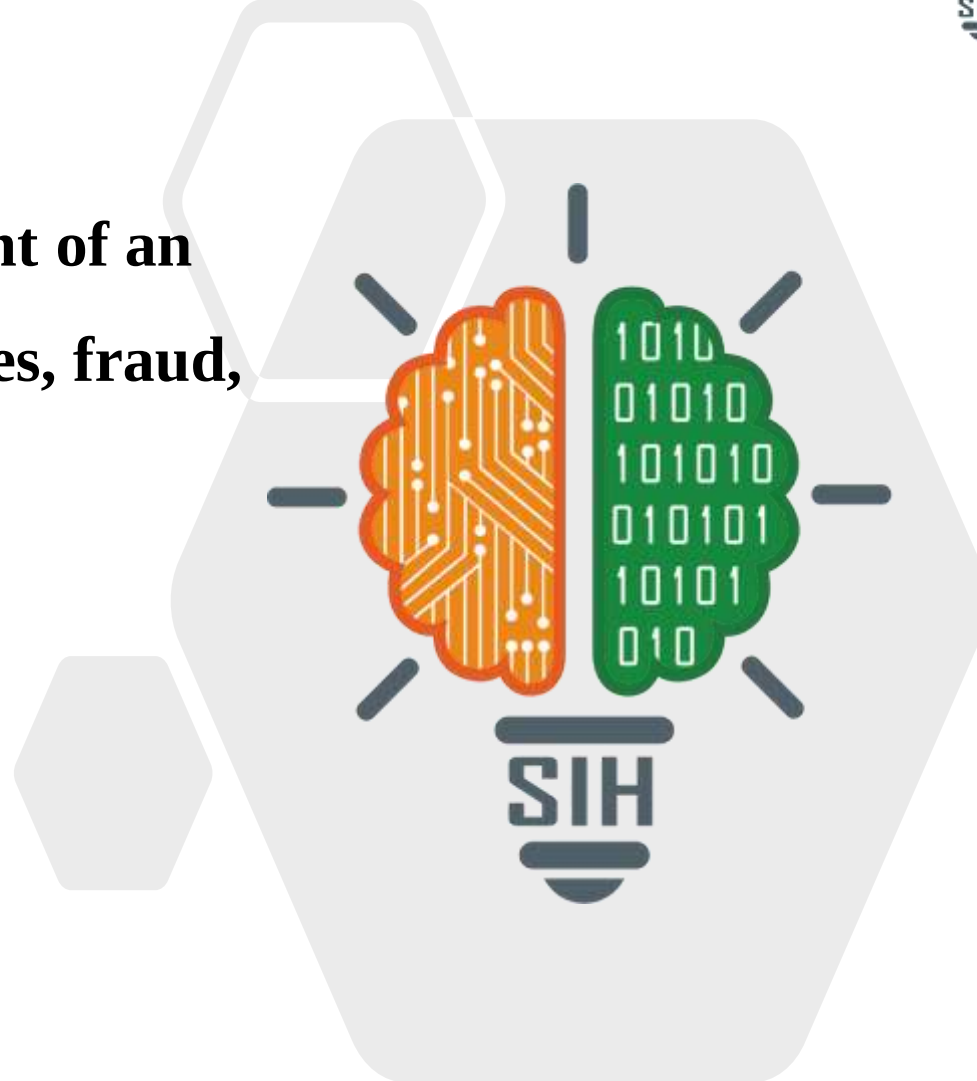


SMART INDIA HACKATHON 2026



- **Problem Statement ID- SIH26102**
- **Problem Statement Title- Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID- MITS/SIH26/097**
- **Team Name- Shatkon**



AI-Assisted Anomaly Detection & Risk Prioritization for MPLAD Monitoring

PROBLEM

- MPLAD generates thousands of work & expenditure records every year across states, MPs, and agencies
- Manual screening for irregularities and inefficiencies does not scale and stays inconsistent.
- Unusual costs, delays & fund-utilization patterns can remain hidden.
- Existing systems record data, but don't intelligently prioritize risk.

OUR SOLUTION

- Connect MPLAD data using Work_ID
- Analyze cost, delay and fund utilization
- Compare patterns across different levels
- Detect unusual patterns using anomaly detection.
- Generate anomaly scores with explainable reasons.
- Prioritize high-risk cases for human review

INNOVATION & UNIQUENESS

MULTI-LEVEL PATTERN ANALYSIS

Analyzes anomalies across Work, Agency, Constituency and Time — not just individual transactions.

EXPLAINABLE ANOMALY PRIORITIZATION

Combines anomaly scores with human-readable reasons to prioritize cases for official review.

HUMAN-IN-THE-LOOP MONITORING

Flags unusual patterns for investigation instead of declaring them as confirmed fraud.



TECHNICAL APPROACH

1. MPLAD DATA SOURCES

Works, expenditure and related records

2. DATA CLEANING & INTEGRATION

Deduplication, validation, missing values and Work_ID-based merging

3. STRUCTURED DATA STORAGE

PostgreSQL for cleaned data and derived features

4. FEATURE ENGINEERING

Cost, utilization, duration, frequency and temporal patterns

5. ANOMALY DETECTION

Isolation Forest identifies unusual records/patterns

6. ANOMALY SCORES & EXPLANATIONS

Score + human-readable reasons for flagged cases

7. FASTAPI SERVICE LAYER

Serves processed data and model outputs

8. REACT+TYPESCRIPT DASHBOARD

Filtering, visualization and anomaly review

9. HUMAN REVIEW

Officials investigate flagged cases and take action

TECH STACK:

- Frontend: React.js+Typescript — interactive monitoring dashboard
- Backend: FastAPI — API and model integration
- Data Processing: Python + Pandas — cleaning, integration & feature engineering
- Machine Learning: Scikit-learn + Isolation Forest — anomaly detection
- Database: PostgreSQL — store cleaned data, features & anomaly results
- Data Sources: MPLADS / eSAKSHI official records + government datasets

RISK ASSESSMENT & MITIGATION

PROBLEM: Incomplete & inconsistent MPLAD datasets



Mitigation:
Data cleaning, validation & missing-value handling

PROBLEM: Context varies across works



Mitigation:
Peer/context-aware comparison

PROBLEM: Lack of trust in AI results



Mitigation:
Human-readable anomaly explanations

PROBLEM: Multiple anomaly signals



Mitigation:
Multi-signal risk scoring

TECHNICAL FEASIBILITY

- Python + Pandas for data cleaning & integration.
- PostgreSQL for structured & scalable storage.
- Scikit-learn Isolation Forest for unsupervised anomaly detection.
- FastAPI for REST API integration.
- React dashboard for visualization & monitoring.
- Explainable anomaly scores for human verification.

ECONOMIC & OPERATIONAL VIABILITY

- Low-cost: Uses open-source technologies.
- Scalable: Can process large numbers of MPLAD records.
- Leverages existing data: Reuses existing government data.
- Integration : Integrates with existing monitoring

IMPACT AND BENEFITS



IMPACTS:

- **FINANCIAL ACCOUNTABILITY:** Strengthens oversight of public expenditure and unusual financial patterns..
- **ADMINISTRATIVE GOVERNANCE:** Enables a shift from broad manual scrutiny to risk-based monitoring.
- **TRANSPARENCY:** Creates a consistent, evidence-oriented basis for reviewing works.
- **PUBLIC RESOURCE MANAGEMENT:** Supports better oversight of fund utilization and project implementation.
- **DIGITAL GOVERNANCE:** Introduces AI-assisted analytics into large-scale scheme monitoring.

BENEFITS:

- **UNIFIED ANALYTICAL VIEW:** Brings work, sanction, expenditure, agency and progress information together.
- **RISK-BASED PRIORITY QUEUE:** Ranks works into risk levels for structured review.
- **EXPLAINABLE CASE ALERTS:** Shows the key signals behind why a work was flagged.
- **PEER & COMPARATIVE ANALYSIS:** Enables comparison with relevant similar works and contextual patterns.
- **INVESTIGATION SUPPORT:** Provides supporting work and expenditure details for human-led verification.

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